# Tutorial 1: First-Order ODEs (Part I)

[Attempt all questions before tutorial session]

## A) Types of ODE

For each of the following ODE determine its order, and whether it is linear or non-linear.

a) $x^2\dfrac{d^2y}{dx^2} + 3x\dfrac{dy}{dx} + 2y = \sin x$ b) $\left(4+y^2\right)\dfrac{d^2y}{dt^2} + t\dfrac{dy}{dt} + y = e^t$

c) $\dfrac{d^4y}{dx^4} + \dfrac{d^3y}{dx^3} + \dfrac{d^2y}{dx^2} + \dfrac{dy}{dx} + y = 2$ d) $\dfrac{dy}{dt} + 2ty^2 = 0$

e) $\dfrac{d^2y}{dx^2} + \sin(x+y) = 2\sin x$ f) $\dfrac{d^3y}{dt^3} + t\dfrac{dy}{dt} + (\cos t)^2 y = 4t^3$

**Solution:** d): first order; a), b), e): second order; f): third order; c) fourth order

Linear: a), c), f); Nonlinear: b), d), e)

## B) Explicit vs implicit solutions

Verify that the indicated function $y$ is a solution of the given differential equation.

a) $\dfrac{dy}{dx} + 2xy = 1; \quad y = e^{-x^2}\displaystyle\int_0^x e^{t^2}\,dt + ce^{-x^2}.$

b) $2y' = y^3\cos x; \quad y = (1-\sin x)^{-1/2}.$

c) $y' = \dfrac{y^2}{1-xy}; \quad xy = \ln y + c.$

**Solution:** a) Recall the differentiation formula

$$\frac{d}{dx}\int_a^x f(t)dt = f(x).$$

Then we take derivative on

$$\begin{aligned}
\frac{dy}{dx} &= \frac{d}{dx}\Big(e^{-x^2}\int_0^x e^{t^2}dt + ce^{-x^2}\Big) \\
&= \frac{d}{dx}(e^{-x^2})\int_0^x e^{t^2}dt + e^{-x^2}\frac{d}{dx}\int_0^x e^{t^2}dt + c\frac{d}{dx}(e^{-x^2}) \\
&= -2xe^{-x^2}\int_0^x e^{t^2}dt + 1 - 2cxe^{-x^2} \\
&= -2xe^{-x^2}\Big(\int_0^x e^{t^2}dt + c\Big) + 1 = -2xy + 1.
\end{aligned}$$

b) We rewrite

$$y = (1-\sin x)^{-1/2} \quad \Rightarrow \quad (1-\sin x)y^2 = 1.$$

Then we differentiate it with respect to $x$ and find

$$\begin{aligned}
&\frac{d}{dx}\big\{(1-\sin x)y^2\big\} = 0 \quad \Rightarrow \quad -(\cos x)y^2 + (1-\sin x)2yy' = 0 \\
&\Rightarrow \quad y' = \frac{(\cos x)y}{2(1-\sin x)} = \frac{1}{2}(\cos x)y^3 \quad \Rightarrow \quad 2y' = y^3\cos x.
\end{aligned}$$

c) By the implicit differentiation or chain rule, we have

$$\begin{aligned}
&(xy)' = (\ln y + c)' \quad \Rightarrow \quad y + xy' = \frac{y'}{y} + 0 \quad \Rightarrow \quad y + xy' = \frac{y'}{y} + 0 \\
&\Rightarrow \quad y' = \frac{y^2}{1-xy}.
\end{aligned}$$

C) **Nonlinear first-order ODE (Separable Equations)**

Solve the following equations:

1) $y' - 8y^3 = 0$

2) $yy' = 3y + 2$

3) $yy' = 4y^2 + 1$

4) $y' = y^2 - 4$

5) $y' = y^2 - 2y$

6) $y' = (x^2 + x + 1)(y^2 + 4y + 4)$

7) $yy' + 4x = 0, \quad y(2) = 3$

**Solution:** 1) It is separable. If $y \neq 0$, we have

$$y' - 8y^3 = 0 \quad \Rightarrow \quad \frac{dy}{dx} = 8y^3 \quad \Rightarrow \quad \int \frac{dy}{y^3} = \int 8\,dx + C$$
$$\Rightarrow \quad -\frac{1}{2}\frac{1}{y^2} = 8x + C \quad \Rightarrow \quad \frac{1}{y^2} = -16x - 2C.$$

It is easy to verify that $y = 0$ is a solution of the equation, so we have the general solution is

$$y^2 = -\frac{1}{16x + 2C} \quad \text{or} \quad y = 0.$$

2) It is separable. If $3y + 2 \neq 0$, then

$$yy' = 3y + 2 \quad \Rightarrow \quad y\frac{dy}{dx} = 3y + 2 \quad \Rightarrow \quad \int \frac{y}{3y+2}dy = \int dx + C$$
$$\Rightarrow \quad \int \Big(\frac{1}{3} - \frac{2}{3}\frac{1}{3y+2}\Big)dy = x + C \quad \Rightarrow \quad \frac{y}{3} - \frac{2}{9}\ln|3y+2| = x + C.$$

We also verify that $y = -2/3$ is a solution which cannot be obtained by taking any $C$. Therefore, the solution is

$$\frac{y}{3} - \frac{2}{9}\ln|3y+2| = x + C \quad \text{or} \quad y = -\frac{2}{3}.$$

3) It is separable:

$$yy' = 4y^2 + 1 \quad \Rightarrow \quad y\frac{dy}{dx} = 4y^2 + 1 \quad \Rightarrow \quad \int \frac{ydy}{4y^2+1} = \int dx + C$$
$$\text{Integrate : } \frac{1}{8}\ln\left|4y^2 + 1\right| = x + C \quad \Rightarrow \quad \ln(4y^2+1) = 8x + 8C$$

The general solution is

$$y^2 = \tilde{C}e^{8x} - \frac{1}{4}.$$

4) It is separable. If $y^2 \neq 4$ (i.e, $y \neq \pm 2$), then

$$y' = y^2 - 4 \Rightarrow \frac{dy}{dx} = y^2 - 4$$
$$\text{Separate variables : } \int \frac{dy}{y^2-4} = \int dx + C \Rightarrow \int \frac{dy}{(y+2)\,(y-2)} = \int dx + C$$
$$\Rightarrow \int \frac{-1/4}{y+2} + \frac{1/4}{y-2}dy = x + C$$
$$\text{Integrate : } -\frac{1}{4}\ln|y+2| + \frac{1}{4}\ln|y-2| = x + C, \quad \text{so} \quad \ln\left|\frac{y-2}{y+2}\right| = 4x + 4C.$$

Taking exponential on both sides leads to

$$\left|\frac{y-2}{y+2}\right| = e^{4C}e^{4x} \quad \Rightarrow \quad \frac{y-2}{y+2} = \pm e^{4C}e^{4x} \quad \Rightarrow \quad y = 2\frac{1+\tilde{C}e^{4x}}{1-\tilde{C}e^{4x}}.$$

It is evident that $y = \pm 2$ are solutions, while we get $y = 2$ by taking $\tilde{C} = 0$. Thus, the general solution is

$$y = 2\frac{1+\tilde{C}e^{4x}}{1-\tilde{C}e^{4x}} \quad \text{or} \quad y = -2.$$

5) can be solved in the same way as 4). The general solution is

$$\frac{1}{y} = \frac{1}{2} - Ce^{2x} \quad \text{or} \quad y = 0.$$

6) It is separable, and if $y \neq -2$,

$$y' = \left(x^2+x+1\right)\left(y^2+4y+4\right) \Rightarrow \frac{dy}{dx} = \left(x^2+x+1\right)(y+2)^2$$

$$\text{Separate variables}: \int \frac{dy}{(y+2)^2} = \int \left(x^2+x+1\right) dx + C$$

$$\text{Integrate}: -\frac{1}{y+2} = \frac{1}{3}x^3 + \frac{1}{2}x^2 + x + C \Rightarrow y = -\frac{1}{\frac{1}{3}x^3 + \frac{1}{2}x^2 + x + C} - 2$$

Note that $y = -2$ is a solution, but can not be obtained from the above. Thus, the solution is

$$y = -\frac{1}{\frac{1}{3}x^3 + \frac{1}{2}x^2 + x + C} - 2 \ \text{ or } \ y = -2.$$

7) It is separable:

$$yy' + 4x = 0; \quad y(2) = 3$$

$$y\frac{dy}{dx} = -4x$$

$$\text{Separate variables}: \int y\,dy = \int -4x\,dx$$

$$\text{Integrate}: \frac{1}{2}y^2 = -2x^2 + C \quad \Rightarrow \quad y^2 = -4x^2 + C_1$$

$$\text{Subst. } y(2) = 3 \text{ into gen. soln.}: 3^2 = -4(2)^2 + C \Rightarrow C = 25$$

$$\text{Hence the solution is}: y^2 = -4x^2 + 25$$

D) **Reduction to separable form**

Solve

$$2xyy' = y^2 - x^2$$

by using the substitution: $y = xV$ (then $V = V(x)$ satisfies a separable DE).

**Solution:** Substitute $y = xV$ inside the equation:

$$2x(xV)(xV)' = (xV)^2 - x^2 \quad \text{(eliminate } x^2\text{)}$$

$$2V(xV)' = V^2 - 1 \quad \Rightarrow \quad 2V(V + xV') = V^2 - 1$$

$$2xVV' = -(V^2 + 1) \qquad \text{(Separable DE)}$$

$$\frac{2V}{V^2+1}dV = -\frac{1}{x}dx$$

$$\text{General Solution: } \int \frac{2V}{V^2+1}dV = -\ln|x| + C$$

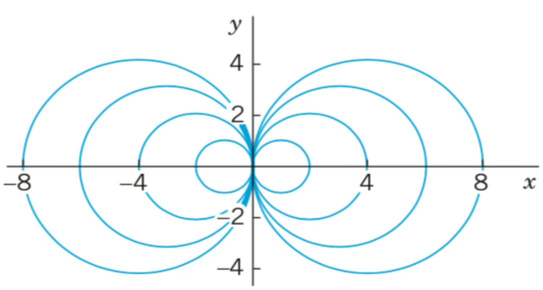
$$\Rightarrow \quad \ln(V^2 + 1) = -\ln|x| + C \quad \text{(Exponentiate it)}$$

$$\Rightarrow \quad V^2 + 1 = \frac{c}{x}$$

Substituting $V = \frac{y}{x}$ and multiplying $y^2$ on both sides, we obtain

$$y^2 + x^2 = cx \quad \text{i.e.,} \quad \left(x - \frac{c}{2}\right)^2 + y^2 = \frac{c^2}{4}.$$

This general solution represents a family of circles passing through the origin with centres on the $x$-axis.



E) **Linear first-order ODE**

Solve the following equations:

$$\begin{aligned}
&1)\ \ y' = 4y + x\\
&2)\ \ x^2y' + 3xy = \frac{1}{x},\ x > 0; \quad y(1) = -1\\
&3)\ \ y' = -2y + 3\\
&4)\ \ y' = y - 1\\
&5)\ \ y' - 4y + 1 = 0
\end{aligned}$$

**Solution:** 1) It is first-order linear DE. We write $y' = 4y + x$ in the standard form with $p(x) = -4, q(x) = x$, and evaluate the integrating factor (IF):

$$y' - 4y = x \quad \Rightarrow \ \text{IF}: \ \ I(x) = e^{\int -4dx} = e^{-4x}.$$

Then by the solution formula, we have

$$\begin{aligned}
y &= \frac{1}{I(x)}\Big\{\int I(x)q(x)dx + C\Big\} = e^{4x}\Big\{\int xe^{-4x}dx + C\Big\}\\
&= -\frac{x}{4} - \frac{1}{16} + Ce^{4x},
\end{aligned}$$

where we used integration by parts to evaluate

$$\begin{aligned}
\int xe^{-4x}dx = -\frac{1}{4}\int x\,d(e^{-4x}) &= -\frac{1}{4}xe^{-4x} + \frac{1}{4}\int e^{-4x}dx\\
&= -\frac{1}{4}xe^{-4x} - \frac{1}{16}e^{-4x}.
\end{aligned}$$

2) We first write the equation in the standard form:

$$x^2y' + 3xy = \frac{1}{x} \quad \Rightarrow \quad \frac{dy}{dx} + \frac{3}{x}y = \frac{1}{x^3} \quad \Rightarrow \quad \text{IF}: \ \ I(x) = e^{\int \frac{3}{x}dx} = x^3,$$

where we take the positive sign for $I(x) = |x|^3 = \pm x^3$. Then by the solution formula, we have

$$y = \frac{1}{I(x)}\Big\{\int I(x)q(x)dx + C\Big\} = \frac{1}{x^3}\Big\{\int x^3\frac{1}{x^3}dx + C\Big\} = \frac{1}{x^2} + \frac{C}{x^3}.$$

Substitute $y(x=1)=-1$ into the equation:

$$-1 = \frac{1}{1^2} + \frac{C}{1^3} \quad \Rightarrow \quad C = -2.$$

The solution is

$$y = \frac{1}{x^2} - \frac{2}{x^3}.$$

3) It can be solved by method for first-order linear DE or separable DE. Here, we treat it as a separable DE:

$$y' = -2y + 3 \quad \Rightarrow \quad \frac{dy}{-2y+3} = dx \quad \Rightarrow \quad -\frac{1}{2}\ln|-2y+3| = x + C$$
$$\Rightarrow \quad \ln|3-2y| = -2x - 2C \quad \Rightarrow \quad 3 - 2y = \pm e^{-2C}e^{-2x} = \tilde{C}e^{-2x}.$$

Thus, the general solution is

$$y = \frac{3}{2} - \frac{\tilde{C}}{2}e^{-2x}.$$

4)-5) can be solved as 3), so we just provide the answers: 4): $y = Ce^x + 1$; 5) $y = \frac{1}{4} + Ce^{4x}$.

F) **Modeling by ODE**

The mixing of two salt solutions of different concentrations gives rise to a first-order differential equation for the amount of salt contained in the mixture. Let us suppose that a large mixing tank initially holds 300 gallons of brine (that is, water in which a certain number of pounds of salt has been dissolved). Another brine solution is pumped into the large tank at a rate of 3 gallons per minute; the concentration of the salt in this inflow is 2 pounds per gallon. When the solution in the tank is well stirred, it is pumped out at the same rate as the entering solution. Let $A(t)$ denote the amount of salt in the tank at time t. Find the differential equation that $A$ satisfies, and solve it.

**Solution:** Let $A(t)$ be the amount of salt at time $t$. Then the rate of change of $A(t)$

can be equated as

$$\begin{aligned}\frac{dA}{dt} &= \big(\text{Input rate of salt}\big) - \big(\text{output rate of salt}\big)\\ &= \big(2\ \text{lb/gal}\big)\cdot\big(3\ \text{gal/min}\big) - \Big(\frac{A(t)}{300}\ \text{lb/gal}\Big)\cdot\big(3\ \text{gal/min}\big)\\ &= \big(6\ \text{lb/min}\big) - \Big(\frac{A(t)}{100}\ \text{lb/min}\Big).\end{aligned}$$

This gives the differential equation

$$\frac{dA}{dt} = 6 - \frac{A}{100},$$

which is a linear DE with the general solution:

$$A(t) = 600 + ce^{-t/100}.$$